WILLIAM LEE

Senior Generative AI Engineer - Multimodal, RAG and Agents

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PROFESSIONAL SUMMARY

Senior Generative AI Engineer - Multimodal, RAG and Agents with 10+ years delivering secure software across application security, device protection, learning technology, and digital media, using Python 3.9-3.12, PyTorch 2.x and modern foundation-model tooling; combines hands-on architecture, modernization, incident leadership, and mentoring to build scalable systems that improve performance, reliability, security, quality, customer outcomes, and release speed.

SKILLS

AI/ML Engineering: Python, PyTorch, multimodal models, RAG, agents, vector databases, evaluation, safety and model gateways
Data & Pipelines: PostgreSQL, object storage, Kafka, Spark, Airflow, dbt, data validation, feature pipelines, batch and streaming processing
MLOps & Cloud: MLflow, model registries, Docker, Kubernetes, Terraform, CI/CD, managed AI services on AWS, Azure and GCP
APIs & Applications: FastAPI, Django/Flask, Node.js, REST, GraphQL, gRPC, event-driven integration, authentication and rate limiting
Quality & Observability: offline and online evaluation, data and model drift, tracing, prompt/model telemetry, performance tests, security and privacy controls
Architecture & Leadership: responsible AI, governed model access, feature stores, model gateways, Architecture reviews, technical discovery, estimation, mentoring, code review, incident leadership, Agile delivery, stakeholder communication, roadmap planning

WORK EXPERIENCE

Self-Employed / Independent Consulting | Jun 2026 - Present | Tennessee, United States

Staff Software Engineer

- Architected and delivered **multimodal assistants, retrieval pipelines and production agent workflows** using **Python 3.11-3.12, multimodal models, RAG, agents and evaluation**, establishing modular boundaries, API contracts, secure defaults, and an incremental roadmap that balanced near-term releases with long-term maintainability.
- Modernized legacy services and user flows into **model gateways, hybrid retrieval, multimodal pipelines and policy-aware agent orchestration**, resolving memory leaks, N+1 database queries, race conditions, stale-cache defects, and timeout failures to reduce **p95 response time by 38%**.
- Built automated delivery on **cloud services** with containers, infrastructure as code, environment promotion, secrets management, and rollback controls, shortening **release lead time by 42%** without weakening auditability.
- Designed secure **REST, GraphQL, gRPC, webhook, and event-driven integrations** with OAuth 2.0, OpenID Connect, short-lived credentials, idempotency, rate limiting, and contract-based compatibility controls.
- Introduced unit, integration, contract, end-to-end, performance, and security testing around **Python, multimodal models, RAG, agents and MLOps**, reducing escaped defects by **31%** and making production releases more predictable across remote teams.
- Optimized relational queries, cache strategy, search indexes, asynchronous workloads, and cloud resource sizing, cutting recurring infrastructure cost by **24%** while preserving availability and growth headroom.
- Led architecture reviews, technical discovery, estimation, code reviews, and incident retrospectives, mentoring engineers through practical design decisions rather than relying on framework-first or rewrite-first solutions.
- Implemented observability with structured logs, metrics, distributed traces, SLO dashboards, and actionable alerts, improving fault isolation and supporting a sustained **99.95% service availability** target.
- Partnered with product, security, data, and operations stakeholders to convert ambiguous requirements into sequenced technical outcomes, documented ADRs, migration plans, and measurable business acceptance criteria.

Veracode | Jul 2021 - May 2026 | Somerville, MA

Staff Software Engineer

- Led architecture and hands-on implementation for **developer security assistants, multimodal artifact analysis and retrieval-grounded remediation** using **Python 3.9-3.11, transformer models, vector search and model-serving foundations**, supporting multi-tenant SaaS growth while maintaining tenant isolation, secure coding controls, and backward-compatible customer workflows.
- Refactored tightly coupled modules into **model gateways, hybrid retrieval, multimodal pipelines and policy-aware agent orchestration**, eliminating duplicate processing, connection-pool exhaustion, queue backlogs, and inconsistent retries to improve sustained processing throughput by **34%**.
- Designed versioned APIs and asynchronous contracts for findings, policies, scans, notifications, and external integrations, applying idempotency keys, schema evolution, circuit breakers, and dead-letter recovery.
- Modernized legacy components through staged migrations, compatibility adapters, feature flags, and dual-run validation, avoiding a disruptive rewrite while steadily retiring unsupported libraries and build tooling.
- Strengthened quality gates with static analysis, dependency scanning, unit and integration suites, consumer-driven contract tests, and targeted load tests, reducing escaped production defects by **29%**.
- Improved deployment automation across **cloud services** with immutable artifacts, environment policy checks, progressive rollout, and automated rollback, reducing median change lead time by **40%**.
- Diagnosed high-severity production issues involving malformed scan payloads, cross-service timeouts, duplicate events, authorization edge cases, and noisy alerts, then converted findings into durable platform fixes and runbooks.
- Mentored senior and mid-level engineers, facilitated design reviews, and aligned product, security, support, and data teams on technical tradeoffs, service ownership, operational readiness, and measurable delivery outcomes.

Asurion | May 2018 - Jun 2021 | Nashville, TN

Senior Software Engineer

- Developed and evolved **guided customer-service and claim-assistance experiences grounded in policy and device data** with **Python 3.6-3.8, NLP and image-model experimentation with managed ML services**, designing clear service boundaries and reusable components for customer journeys that had to remain responsive during seasonal and partner-driven traffic spikes.
- Reworked synchronous claim and diagnostic paths into asynchronous, retry-safe workflows with caching and database indexing, reducing average transaction processing time by **27%** and improving peak-load stability.
- Resolved recurring production failures caused by duplicate callbacks, partial partner responses, stale device state, slow queries, and inconsistent error mapping, adding idempotency, reconciliation jobs, and clearer operational telemetry.
- Introduced API versioning, schema validation, OAuth-based authorization, secure secret handling, and contract tests for internal and partner integrations, reducing integration regressions and support escalation volume.
- Expanded automated test coverage across unit, integration, browser or device, and performance scenarios, using CI quality gates and representative test data to reduce customer-reported software defects by **22%**.
- Supported legacy framework and runtime upgrades through dependency audits, compatibility layers, canary releases, and rollback plans, enabling modernization without interrupting claims, diagnostics, or fulfillment operations.
- Collaborated with product, UX, operations, and partner teams in Agile delivery, reviewed designs and pull requests, coached engineers, and translated customer-impacting incidents into prioritized reliability improvements.

Rustici Software | Feb 2016 - Mar 2018 | Nashville, TN

Software Engineer

- Built and maintained **content metadata enrichment and retrieval foundations for learning-technology workflows** using **Python 2.7/3.4-3.6, information retrieval and classical machine learning**, translating SCORM, xAPI, cmi5, and LMS interoperability requirements into maintainable application modules and testable integration contracts.
- Diagnosed content-launch failures involving malformed manifests, browser security restrictions, session expiration, inconsistent completion state, and vendor-specific payloads, adding validation, compatibility handling, and clearer diagnostics.
- Improved reporting and content-processing performance through query tuning, batch pagination, background workers, cache controls, and selective denormalization while preserving standards compliance and customer data integrity.
- Modernized legacy modules incrementally with automated regression tests, dependency updates, API adapters, and documented migration paths, avoiding broad rewrites that would have increased customer integration risk.
- Implemented REST integrations, webhooks, message-based processing, authentication controls, and structured logging for customer-hosted and SaaS deployments, improving supportability across varied enterprise environments.
- Worked closely with product specialists, support engineers, and customer developers to reproduce difficult interoperability defects, review code, document behavior, and deliver fixes with clear upgrade guidance.

Mediacube | Jun 2015 - Jan 2016 | Knoxville, TN

Junior Software Engineer

- Implemented features for **media metadata and content-processing foundations that later supported AI-assisted workflows** using **Python 2.7/3.3, text analytics and media-processing pipelines**, converting product requirements into maintainable application code, SQL changes, background jobs, and user-facing workflows under senior engineering guidance.
- Fixed defects involving duplicate campaign records, failed media uploads, incorrect time-zone calculations, inconsistent totals, and slow reporting queries, adding validation, indexes, transaction boundaries, and actionable logging.
- Created and consumed REST endpoints, integrated third-party media and reporting services, and added retry and error-handling behavior for unreliable external responses and scheduled processing jobs.
- Expanded unit and integration checks, participated in code reviews, documented deployment steps, and supported Linux-based production releases while learning disciplined debugging and change-management practices.
- Collaborated with designers, account teams, and experienced engineers to break work into testable increments, clarify edge cases, and deliver digital-media features without disrupting existing customer reporting.

SELECTED PROJECTS

Multimodal RAG and Agent Platform

Designed a production reference system with **Python, multimodal models, RAG, agents and MLOps**, governed data ingestion, evaluation gates, secure model access, versioned APIs, observability, containerized deployment, and infrastructure as code across AWS, Azure, or GCP.

Continuous Evaluation and Operations Framework

Built automated quality, latency, cost, safety, drift, and regression checks with experiment tracking, release approval criteria, dashboards, and rollback workflows, enabling repeatable promotion from prototype to monitored production service.

EDUCATION

University of Tennessee, Knoxville | Sep 2011 - May 2013

Bachelor of Science, Computer Science